Sean Lumasag

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Education

Rutgers University — New Brunswick, NJ
B.S. in Computer Science — *Expected December 2027*

- GPA: 3.90 | Coursework: Data Structures & Algorithms, Database Systems, Computer Architecture

Experience

Software Engineer Intern — May 2026 – Present
Heller Industries — *Florham Park, NJ*

- Implemented asynchronous oven anomaly detection in C#/.NET by applying 8 Nelson rules to heater and conveyor sensor data, identifying abnormal equipment patterns for dashboard investigation
- Built an oven setup importer by validating JSON files and deriving heater and sensor geometry, reconstructing application settings for multiple oven configurations without manual reentry
- Developed thermal-profile comparison software by interpolating simulated curves at measured sampling times, quantifying temperature differences between physical and virtual profiles

Machine Learning Research Assistant — January 2026 – May 2026
WINLAB, Rutgers University — *Piscataway, NJ*

- Automated local-versus-edge inference evaluations by scripting Python/PyTorch experiments across model-routing configurations, producing comparable accuracy and offloading results
- Benchmarked trained against random inference routing by evaluating 10,000 CIFAR-10 images at comparable offloading rates, demonstrating 92.42% versus 89.10% accuracy

Software Engineer Intern — May 2025 – August 2025
Data Storytelling — *Milltown, NJ*

- Extended raffle checkout in TypeScript by persisting application transactions in Supabase after PayPal capture and linking their IDs to tickets, blocking ticket issuance when transaction storage fails
- Implemented a raffle ticket checker by matching submitted ticket numbers against stored Supabase winners, preserving access to drawing results across page reloads and return visits
- Resolved missing organizer raffles by correcting mismatched ownership identifiers in Supabase queries, restoring dashboard access to raffles created by signed-in organizations

Projects

URL Shortener | *Java, Spring Boot, PostgreSQL, Redis*

- Created a Spring Boot URL shortener by securing owner-scoped link-management APIs with JWT authentication, supporting custom short links and per-user click analytics for registered users
- Cached URL redirects by checking Redis for destinations before falling back to PostgreSQL, avoiding repeat database lookups while retaining durable click-event analytics

Image Processor | *TypeScript, NestJS, BullMQ, Redis, PostgreSQL*

- Built an asynchronous image-processing pipeline by dispatching resize and WebP jobs to Redis-backed BullMQ workers, decoupling API requests from image conversions
- Implemented per-image retry handling by configuring 3-attempt exponential backoff in BullMQ and persisting states in PostgreSQL, isolating failures without reprocessing entire batches

Live Chat | *Java, Spring Boot, React, PostgreSQL, WebSockets*

- Built a real-time chat platform by publishing STOMP/WebSocket messages to subscribed conversation rooms, synchronizing direct, group, and public chats across connected clients
- Enforced persist-before-broadcast messaging by committing chat messages through JPA before publishing WebSocket updates, preserving conversation history after restarts

Technical Skills

Languages: Java, C#, Python, TypeScript, SQL
Frameworks: Spring Boot, .NET Framework, NestJS, React
Tools: PostgreSQL, Redis, BullMQ, Git, Docker